26.2: Phylogenies are inferred from morphological and molecular data

To infer phylogeny, systematists must gather as much information as possible about the morphology, genes, and biochemistry of the relevant organisms. It is important to focus on features that result from common ancestry because only those features reflect evolutionary relationships.

Morphological and Molecular Homologies

Recall that phenotypic and genetic similarities due to shared ancestry are called **homologies**. For example, the similarity in the number and arrangement of bones in the forelimbs of mammals is due to their descent from a common ancestor with the same bone structure; this is an example of a morphological homology (see Figure 22.15). In the same way, genes or other DNA sequences are homologous if they are descended from sequences carried by a common ancestor.

In general, organisms that share very similar morphologies or similar DNA sequences are likely to be more closely related than organisms with vastly different structures or sequences. In some cases, however, the morphological divergence between related species can be great and their genetic divergence small (or vice versa). Consider the Hawaiian silversword plants: some of these species are tall, twiggy trees, while others are dense, ground-hugging shrubs (see Figure 25.23). But despite these striking phenotypic differences, the silverswords' genes are very similar. Based on these small molecular divergences, scientists estimate that the silversword group began to diverge 5 million years ago. We'll discuss how scientists use molecular data to estimate such divergence times later in this chapter.

Sorting Homology from Analogy

A potential source of confusion in constructing a phylogeny is similarity between organisms that is due to convergent evolution—called **analogy**—rather than to shared ancestry (homology). Convergent evolution occurs when similar environmental pressures and natural selection produce similar (analogous) adaptations in organisms from different evolutionary lineages. For example, the two mole-like animals shown in **Figure 26.7** look very similar. However, their internal anatomy, physiology, and reproductive systems are very dissimilar. Indeed, genetic and fossil evidence indicate that the common ancestor of these animals lived 160 million years ago. This common ancestor and most of its descendants were not mole-like. It appears that

analogous characteristics evolved independently in these two lineages as they became adapted to similar lifestyles—hence, the similar mole-like features of these animals should not be considered when reconstructing their phylogeny.

Figure 26.7

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Convergent evolution in burrowers.
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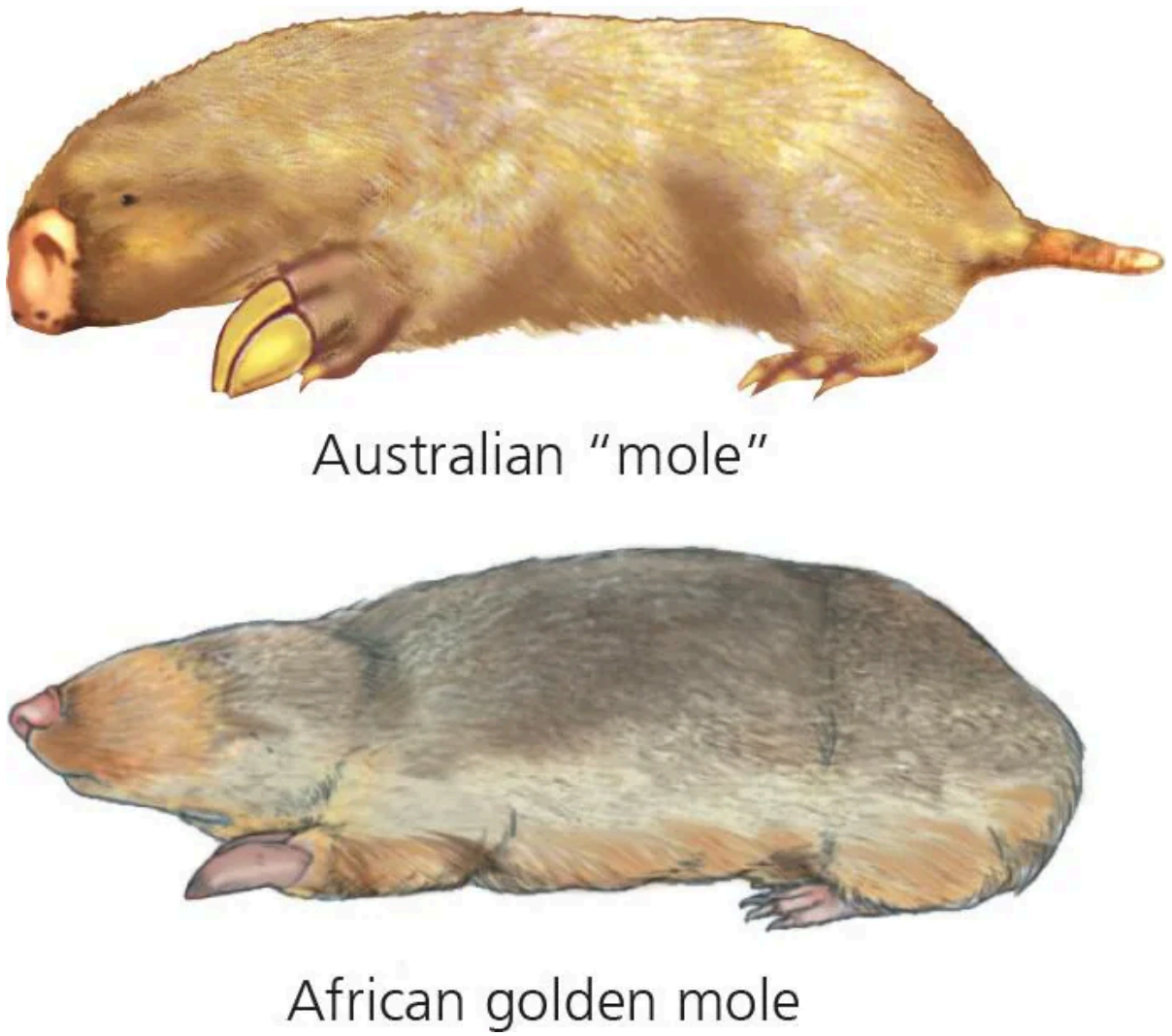

A long body, large front paws, small eyes, and a pad of thick skin that protects the nose all evolved independently in these species.

Another clue to distinguishing between homology and analogy is the complexity of the characters being compared. The more elements that are similar in two complex structures, the more likely it is that the structures evolved from a common ancestor. For instance, the skulls of an adult human and an adult chimpanzee both consist of many bones fused together. The compositions of the skulls match almost

perfectly, bone for bone. It is highly improbable that such complex structures, matching in so many details, have separate origins. More likely, the genes involved in the development of both skulls were inherited from a common ancestor.

The same argument applies to comparisons at the gene level. Genes are sequences of thousands of nucleotides, each of which represents an inherited character in the form of one of the four DNA bases: A (adenine), G (guanine), C (cytosine), or T (thymine). If genes in two organisms share many portions of their nucleotide sequences, it is likely that the genes are homologous.

Evaluating Molecular Homologies

Comparing DNA molecules often poses technical challenges for researchers. The first step after sequencing the DNA is to align comparable sequences from the species being studied. If the species are very closely related, the sequences probably differ at only one or a few sites. In contrast, comparable nucleic acid sequences in distantly related species usually have different bases at many sites and may have different lengths. This is because insertions and deletions accumulate over long periods of time.

Suppose, for example, that certain noncoding DNA sequences near a particular gene are very similar in two species, except that the first base of the sequence has been deleted in one of the species. The effect is that the remaining sequence shifts back one notch. A comparison of the two sequences that does not take this deletion into account would overlook what in fact is a very good match. Computer programs can be used to identify such matches by testing possible alignments for comparable DNA segments of differing lengths **(Figure 26.8)**.

Figure 26.8

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Aligning segments of DNA.
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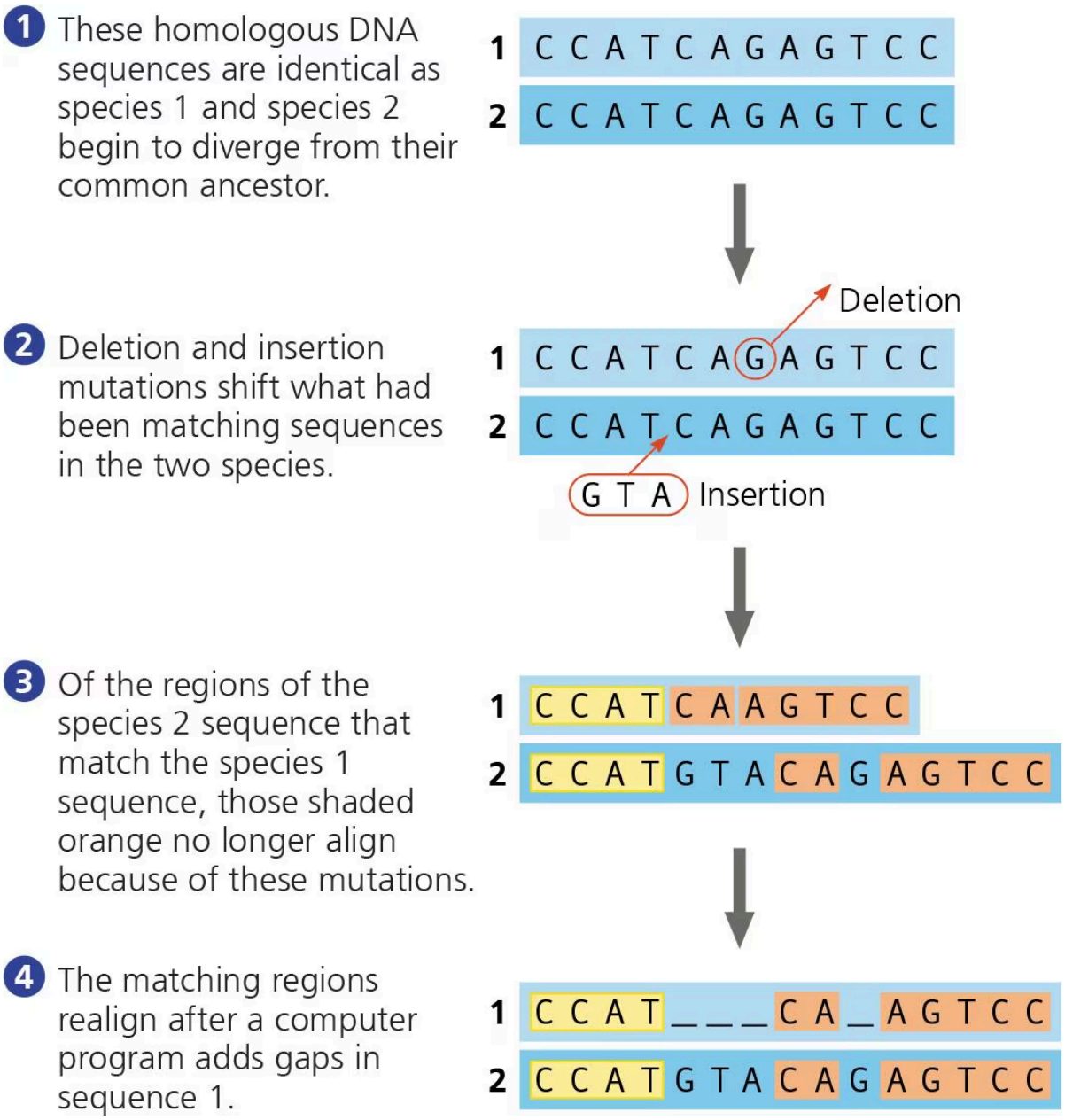

Steps are listed as follows.

1. Homologous DNA sequences are identical as species 1 and species 2 begin to diverge from their common ancestor. Sequence 1. C,C, A, T, C, A, G, A, G, T, C. Sequence 2. C, C, A, T, C, A, G, A, G, T, C.
2. Deletion and insertion mutations shift what had been matching sequences in the two species. In Sequence 1, G is deleted and in Sequence 2, G T A is inserted between T and C.
3. Of the regions of the species 2 sequence that match the species 1 sequence, only the first 4 sequences (C C A T) align because of these mutations. Remaining regions no longer align because of these mutations.

4. The matching regions realign after a computer program adds gaps in sequence 1 where G T A and G occur in sequence 2.

Systematists search for similar sequences along DNA segments from two species (only one DNA strand is shown for each species). In this example, 11 of the original 12 bases have not changed since the species diverged. Hence, those portions of the sequences still align once the length is adjusted.

Such molecular comparisons reveal that many base substitutions and other differences have accumulated in the comparable genes of an Australian “mole” and a golden mole. The many differences indicate that their lineages have diverged greatly since their common ancestor; thus, we say that the living species are not closely related. In contrast, the high degree of gene sequence similarity among the silversword plants indicates that they are all very closely related, in spite of their considerable morphological differences.

Just as with morphological characters, it is necessary to distinguish homology from analogy in evaluating molecular similarities for evolutionary studies. Two sequences that resemble each other at many points along their length most likely are homologous (see Figure 26.8). But in organisms that do not appear to be closely related, the bases that their otherwise very different sequences happen to share may simply be coincidental matches **(Figure 26.9)**. Scientists have developed statistical tools that can help distinguish “distant” homologies from such coincidental matches in extremely divergent sequences.

Figure 26.9

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A coincidental match.
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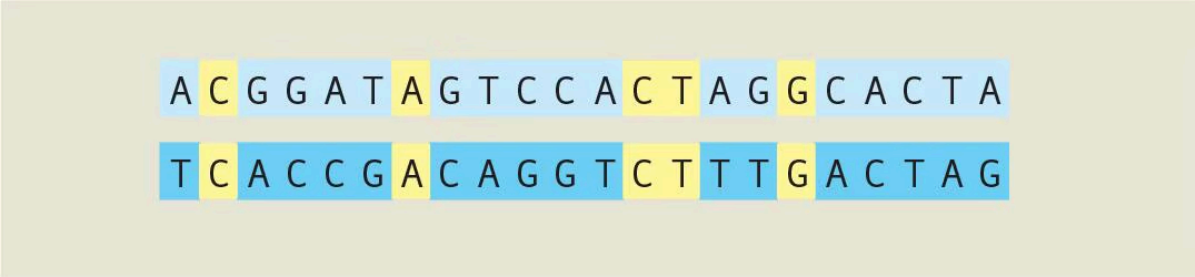

Top sequence: A C G G A T A G T C C A C T A G G C A C T A. Bottom sequence: T C A C C G A C A G G T C T T T G A C T A G. Aligned bases in both the sequences are highlighted in yellow: C (at the second position), A (at seventh position), C T (at the thirteenth and fourteenth positions), and G (at the seventeenth position).

These two DNA sequences from distantly related species coincidentally share 23% of their bases (those highlighted in yellow).

WHAT IF?_Why might you expect organisms that are not closely related to nevertheless share roughly_25%_of their bases?_

*For suggested answer, see*Appendix A.**